Lec 0

Lec-72

algo & gate

Intro of DP

Lec-71

Lec-72

Lec_73

GATE Problems of DP

In GATE :

60-80%.

(done)

Wipro

medium

Competitive coding

Leetcode, GFG, Codechef, Hackerrank

- Fibonacci Series
- Longest common sub sequences (LCS)
- 0/1 knapsack problem
- Sum of subsets
- Matrix chain Multiplication
- Floyd War shall Algorithm

1. fibonacci series

2. Longest common sub sequences (LCS)

A subsequence is a sequence that can be derived from the given string by deleting some or no elements without changing the order of the remaining elements.

S = abcde

For example, "abe" and bd are the subsequences of "abcde"

S_1 = abe

S_2 = bd

Q1: Which are the subsequences of S={ABBABB}

s1=BBBB ✓

s2BABA X

s3=AA ✓

s4=ABAB ✓

s5=BAAB X

S = ABBABB

S_1 = BBBB

S_2 = BABA X

S = ABBABB

S5 = ABBABB

What is the meaning of LCS: Longest common subsequence

- **Our task:Given two strings, s1 and s2, the task is to find the length of the Longest Common Subsequence**

s_1 = ABC

s_2 = ACD

Ex1: Input: s1 = “ABC”, s2 = “ACD”

Output: ? AC Length=2

Explanation:

$S_1 = \{A, B, C, AB, AC, BC, ABC\}$

$S_2 = \{A, C, D, AC, CD, AD, ACD\}$

Ex2:Input: s1 = “AGGTAB”, s2 = “GXTXAYB”

Output: ?

$s_1 =$ A G G T A B

$s_2 =$ G X T X A Y B

length of LCS:

5–6 10

Ans = $\boxed{\text{GTAB}}$ 4

Ex3: x=bababa ,y=abbabaa hoe many LCS are there and also what is the length

m·2

x = bababa
y = abbabaa

S_1 = ababa
S_2 = bbaba
S_3 = baaba
S_4 = babba
S_5 = babaa
S_6 = babab

= 6

S_1 = babaa (5)
S_2 = ababa
S_3 = bbaba

(5)

5

LCS = 3

Nu = 5

x = 10
y = 15
Nu = 7

Ex4:

GATE | GATE-CS-2014-(Set-2) | Question 47

Consider two strings A = "qpqrr" and B = "pqprqrp". Let x be the length of the longest common subsequence (not necessarily contiguous) between A and B and let y be the number of such longest common subsequences between A and B. Then x + 10y = ___.

34

4:15

x = length of LCS

y = no. of LCS

x + 10y = ?

A = qpqrr (5)

B = pqprqrp (7)

s_1 = q, s_2 = qp, s_3 = qpq

$\boxed{s_4 = qpqr}$ ⇒ LCS

length(x) = 4

A = qpqrr

A1 = pqrr ✗

A2 = qqrr ✓

A3 = qpqr ✓

A4 = qpqr ✓

y = 3

4 + 10(3)

⇒ 4 + 30

= 34

How to find LCS?

Using algorithm

Coding